MINOR PROJECT 2: PREDICT NEW YORK CITY AIRBNB PRICES

Course: UNLOX Job Bridge Course — Artificial Intelligence
Project Type: Machine Learning — Regression
Algorithm: Linear Regression

1. Objective

The objective of this project is to develop a machine learning model that predicts Airbnb listing prices in New York City using relevant listing features. Linear Regression is used to learn the relationship between these features and the listing price.

2. Data Exploration and Preprocessing

The dataset was explored using Pandas functions such as `head()`, `tail()`, `info()`, `describe()`, and `isnull().sum()`. Missing values were investigated, missing prices were removed, missing `reviews_per_month` values were filled with zero, and the `last_review` column was dropped. Unnecessary identifier and text columns were also removed.

Potential outliers were examined using boxplots and summary statistics. Prices above the 95th percentile ($350) and listings with minimum stays of 365 nights or more were excluded. The `neighbourhood` column was removed because of its high number of unique categories. One-Hot Encoding was applied to `neighbourhood_group` and `room_type`.

After preprocessing and outlier removal, 21,053 listings remained for modeling.

3. Model Development

The dataset was divided into training and testing sets using an 80:20 split with `random_state=42`. The training set contained 16,842 listings, while the testing set contained 4,211 listings. A Linear Regression model was trained on the training data and used to predict prices for the test data.

4. Model Evaluation

The model was evaluated using three metrics:

- **R^2 Score: 0.4754 — The model explains approximately 47.54% of the variation in Airbnb prices in the test dataset.**
- **Mean Absolute Error (MAE): $36.53 — The average absolute difference between actual and predicted prices was approximately $36.53.**

- **Root Mean Squared Error (RMSE): $49.45 — The root mean squared prediction error was approximately $49.45, with larger errors having a greater influence on this metric.**

Actual-versus-predicted and residual plots were generated to visually examine prediction performance and errors.

5. Conclusion

This project demonstrated the application of data exploration, data cleaning, outlier handling, categorical encoding, and Linear Regression to an Airbnb price prediction problem. The model achieved an R^2 score of 0.4754, providing a useful baseline for further improvement. Future work could explore additional features, alternative regression algorithms, and hyperparameter tuning to improve predictive performance.

Tools and Libraries: Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.